

Alvin Ndungu

Software Engineer · Backend, Full-Stack & ML Research

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PROFESSIONAL SUMMARY

I am a Software Engineer with 3 years of experience spanning across building and shipping production software as well as AI powered products. I have experience in building scalable backend systems, RESTful APIs, mobile, web and desktop applications. I also have experience with RAG pipelines, multi agent systems, LLM driven automation. I enjoy solving practical problems through both AI driven products as well as well engineered traditional software solutions. I am skilled at owning the full product lifecycle from architecture through deployment. I adapt quickly to new technologies, frameworks, and domains, and I actively embrace opportunities to learn the tools and skills needed to deliver value in new roles.

TECHNICAL SKILLS

Languages & Frameworks: Python (Django), Node.js, TypeScript, JavaScript, React, React Native, Rust (Tauri)

ML & Research: Model training & fine-tuning, data sourcing & curation, biosimulation tool setup, scikit-learn, TensorFlow/Keras

Cloud & DevOps: AWS (EC2, S3, Cognito), GCP (Compute Engine, Cloud Storage), Nginx, Unicorn, pm2, Linux Server Administration

Databases & Architecture: MongoDB, MongoEngine, Firebase Realtime DB, Cloud Firestore, Systems Design

Desktop: Tauri (cross-platform desktop apps via Rust + web frontend)

Tools & Methodologies: REST APIs, Git/GitHub, Agile/Scrum, Product Management, Stripe & RevenueCat Integration

App & Project Management: Google Play Console, Apple App Store Connect, KPI tracking, GitHub Scrum boards, Agile/Scrum, Microsoft Teams Planner

Languages: English (fluent), Swahili (native)

PROFESSIONAL EXPERIENCE

Software Engineer & Product Manager · iZola Limited · August 2023 – Present

Mobile & web application development for healthcare platform Xaidi

- Led the end-to-end development of Xaidi, a healthcare platform designed with React Native and Django/Node.js. Scaled this to over 18,000 downloads on both Google Play and The App Store.
- Developed and looked after the RESTful backward service API that integrated Stripe and RevenueCat for complex subscription and payroll logic.
- Set up and handled the production infrastructure on GCP and AWS using Nginx, Unicorn, and PM2.
- Integrated generative AI solutions (e.g., an AI chat assistant) to enhance the engagement and capabilities of the platform.
- Defined key metrics and KPIs for the product and analyzed the data from these to rationalize and prioritize feature releases.
- Developed and maintained APIs and interfaces for web applications built with React and TypeScript, to create a web-based version of the mobile product.
- Performed code review and best practice enforcement to maintain and improve the engineering team's code.

ML Research Engineer · Vitafluence AI · September 2024 – April 2025

Secondment to the core research team; contributed to AI/ML research in healthcare and biosimulation

- Sourced, cleaned, and curated datasets for training and evaluating machine learning models in a healthcare AI context.
- Trained and iterated on ML models, managing experiment tracking and performance evaluation.
- Set up and maintained biosimulation tools as part of the research infrastructure, supporting computational experiments.

- Conducted applied research into AI regulatory compliance and healthcare AI systems, directly informed by EU AI Act requirements.
- Collaborated with the research team to document findings and translate research outputs into actionable engineering recommendations.

AI Automation Freelance(NDA protected) · January – February 2026 ·

Email to action automation system

- Built an AI system that would process the intent of incoming emails and then would automatically execute actions such as creating to do items, calendar events, notes, or reminders without manual entry.
- Implemented user verification checkpoints for lower confidence actions to reduce AI hallucination risk before actions were committed.

AI Systems Integration

Freelance(NDA protected) · January – April 2026 Local, backend free AI assistant

- Co-developed a standalone desktop AI assistant that could run locally with no backend server, operating directly on local file and folder paths.
- The standout feature on this was that the assistant is able to read multiple local files and generate reports, charts, or spreadsheets, and help users query and understand documents without having to upload them.

CERTIFICATIONS

AWS Cloud Practitioner Essentials · AWS · Issued July 2026

Core AWS concepts across compute, storage, networking, security and billing.

EU AI Act Essentials · KI-Campus (Stifterverband) · Issued April 2025

Covers EU AI Act fundamentals, risk classification, stakeholder obligations (Provider, Deployer, National Authority), implementation timelines, and practical compliance scenarios.

ICDL Certificate · Daystar University · Issued August 2019

Covered Microsoft Office suite (Word, Excel, PowerPoint, Outlook), desktop publishing principles, and introductory .NET development for desktop applications.

EDUCATION

B.Sc. Computer Science · Kabarak University - September 2019 – December 2023